

ENVS-193DS_spring-2026_final

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GitHub Repository: [ENVS-193DS Spring 2026 Final](#)

Problem 1

a.

In Part 1, my co-worker used logistic regression. We can tell because the goal is to predict a probability (will an Avocet use this microhabitat or not, which is a yes/no outcome) from a

continuous measurement (distance to the water edge in meters). Logistic regression is the right tool when your response variable is binary like this.

In Part 2, my co-worker used a chi-square test of independence. This is because both variables being compared are categorical — bird species (Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). A chi-square test tells us whether the two categorical variables are associated with each other or not.

b.

Part 1: American Avocets were more likely to be found in microhabitat sites that were closer to the water's edge. This suggests that how far a spot is from the water plays a role in whether or not an Avocet chooses to use it.

Part 2: The three bird species did not use the five habitat types equally. Each species tended to show up more in certain habitats than others, meaning the habitat a bird is found in is related to which species it is.

c.

One additional variable that could be worth looking at is water depth, measured in centimeters at each survey point using a ruler or depth probe. American Avocets are wading birds that forage by sweeping their bill through shallow water, so sites that are too deep or too dry may simply not be usable for feeding, which would directly affect whether or not Avocets choose to use a given microhabitat.

Problem 2

a.

```
nests_clean <- nests |>
  select(case_control,
         height_cm,
         ant_species) |>
  filter(ant_species == "AB" |
         ant_species == "RRB" |
         ant_species == "BBR") |>
  mutate(ant_species = factor(ant_species,
                             levels = c("AB",
```

```

                                "RRB",
                                "BBR")))) |>
mutate(full_name = case_when(
  ant_species == "AB" ~ "Crematogaster sjostedti",
  ant_species == "RRB" ~ "Crematogaster mimosae",
  ant_species == "BBR" ~ "Crematogaster nigriceps"
))
str(nests_clean)

```

```

'data.frame':  270 obs. of  4 variables:
 $ case_control: int  1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm   : num  392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ full_name   : chr  "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae"

```

```

slice_sample(nests_clean, n = 10)

```

	case_control	height_cm	ant_species	full_name
1	1	195.0	RRB	Crematogaster mimosae
2	0	132.0	BBR	Crematogaster nigriceps
3	0	223.5	RRB	Crematogaster mimosae
4	1	152.5	RRB	Crematogaster mimosae
5	0	329.0	RRB	Crematogaster mimosae
6	1	390.0	BBR	Crematogaster nigriceps
7	0	234.5	RRB	Crematogaster mimosae
8	0	97.5	RRB	Crematogaster mimosae
9	0	132.0	RRB	Crematogaster mimosae
10	0	180.0	RRB	Crematogaster mimosae

b.

The 1s and 0s in the case_control column represent whether a given tree contained a bird nest (1) or was a nearby available tree with no nest present (0). This is a binary outcome used to compare the characteristics of trees that birds actually chose to nest in versus trees that were available but unused.

c.

Tree height (height_cm, continuous variable measured in centimeters): As tree height increases, the probability of nest occurrence would likely increase because taller trees may offer more structural complexity and concealment, making them more attractive and safer nesting locations. The authors note that all nests found in the study were located in live trees more than 1.5 m in height, suggesting a minimum height threshold for nest use. (Methods, paragraph 2; Results, paragraph 1)

Acacia ant species (ant_species, categorical variable with 3 levels: *C. sjostedti*, *C. mimosae*, and *C. nigriceps*): The ant species occupying a tree could influence the probability of nest occurrence because different species vary in how aggressively they defend their host trees against predators. *C. mimosae* and *C. nigriceps* are aggressive defenders against both mammalian and insect herbivores, and birds nesting in those trees may benefit from protection against nest predators. In contrast, *C. sjostedti* provides only minimal protection, making those trees potentially less safe or attractive for nesting birds. Additionally, *C. nigriceps* prunes the apical buds of its host tree, creating a denser canopy that likely provides extra concealment from predators such as snakes, mesocarnivores, and raptors. (Introduction, paragraphs 2 and 4; Discussion, paragraphs 2 and 3)

d.

Model	
Number	Description
Model 1	Probability of nest occurrence predicted by tree height and acacia ant species, no interaction
Model 2	Probability of nest occurrence predicted by tree height and acacia ant species, with interaction between tree height and acacia ant species

e.

```
# Model 1: no interaction
model1 <- glm(case_control ~ height_cm + ant_species,
              data = nests_clean,
              family = binomial)

# Model 2: with interaction
model2 <- glm(case_control ~ height_cm * ant_species,
              data = nests_clean,
              family = binomial)
```

f.

```
AIC(model1, model2)
```

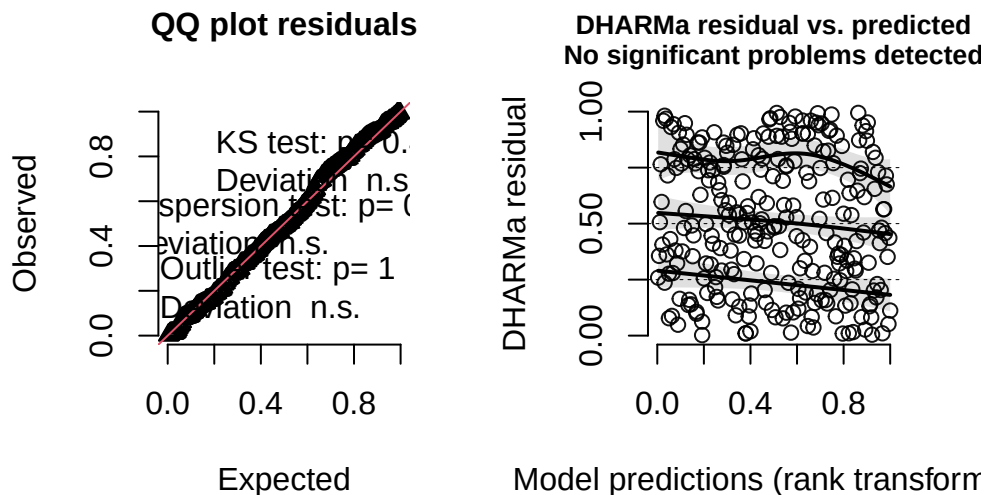
	df	AIC
model1	4	258.7430
model2	6	237.8042

The best model as determined by Akaike's Information Criterion (AIC) was the model predicting probability of bird nest occurrence from tree height and acacia ant species with an interaction term (AIC = 237.80), as it had a lower AIC value than the model without an interaction term (AIC = 258.74).

g.

```
# swap model1 or model2 for whichever was selected in part f
simulated_residuais <- simulateResiduals(fittedModel = model2, plot = TRUE)
```

DHARMa residual



h.

```
# create a grid of values to predict over
pred_data <- expand.grid(
  height_cm = seq(min(nests_clean$height_cm),
                  max(nests_clean$height_cm),
                  length.out = 100),
  ant_species = factor(c("AB", "RRB", "BBR"),
                      levels = c("AB", "RRB", "BBR"))
)

# get predictions and confidence intervals
preds <- predict(model2, newdata = pred_data, type = "link", se.fit = TRUE)

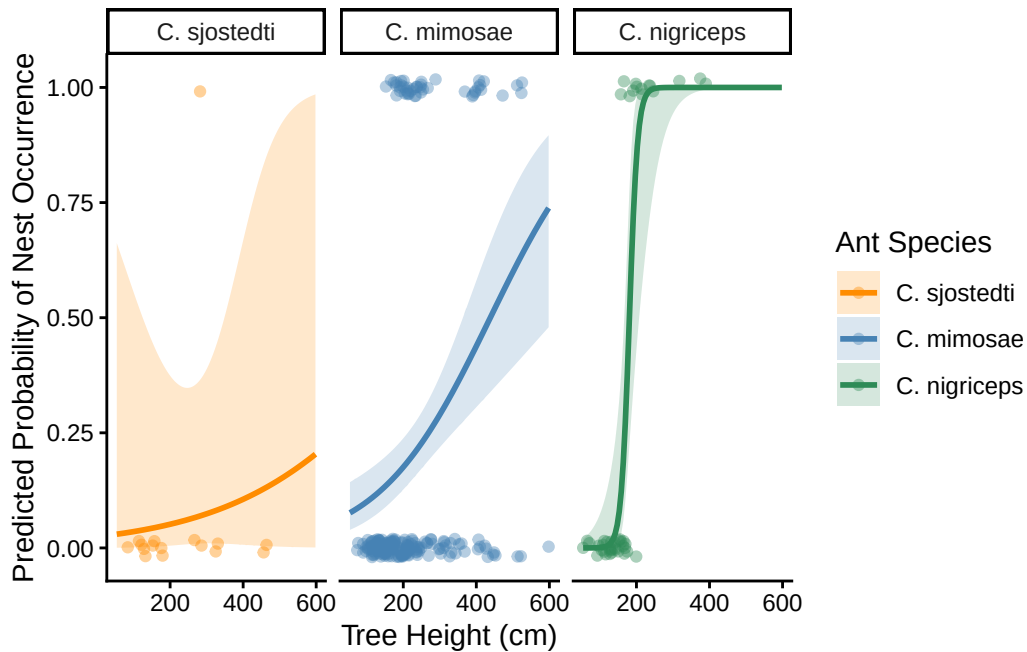
pred_data$fit <- plogis(preds$fit)
pred_data$lower <- plogis(preds$fit - 1.96 * preds$se.fit)
pred_data$upper <- plogis(preds$fit + 1.96 * preds$se.fit)

# plot
ggplot() +
  # raw data points
  geom_jitter(data = nests_clean,
             aes(x = height_cm, y = case_control, color = ant_species),
             height = 0.02, alpha = 0.4) +
  # model prediction lines
  geom_line(data = pred_data,
           aes(x = height_cm, y = fit, color = ant_species),
           linewidth = 1) +
  # confidence interval ribbons
  geom_ribbon(data = pred_data,
            aes(x = height_cm, ymin = lower, ymax = upper, fill = ant_species),
            alpha = 0.2) +
  # one panel per ant species
  facet_wrap(~ ant_species,
            labeller = labeller(ant_species = c("AB" = "C. sjostedti",
                                                "RRB" = "C. mimosae",
                                                "BBR" = "C. nigriceps")))) +
  theme_classic() +
  scale_color_manual(values = c("AB" = "darkorange",
                                "RRB" = "steelblue",
                                "BBR" = "seagreen"),
                    labels = c("C. sjostedti", "C. mimosae", "C. nigriceps")) +
```

```

scale_fill_manual(values = c("AB" = "darkorange",
                             "RRB" = "steelblue",
                             "BBR" = "seagreen"),
                  labels = c("C. sjostedti",
                             "C. mimosae",
                             "C. nigriceps")) +
labs(x = "Tree Height (cm)",
     y = "Predicted Probability of Nest Occurrence",
     color = "Ant Species",
     fill = "Ant Species")

```



i.

Figure 1. Predicted probability of bird nest occurrence by tree height for three acacia ant species. Each panel shows how the predicted probability of a bird nesting in a tree changes as tree height increases, with one panel for each ant species (*Crematogaster sjostedti*, *C. mimosae*, and *C. nigriceps*). The line shows the model's predicted probability and the shaded area around it represents the 95% confidence interval. Individual points show whether a nest was actually present (1) or absent (0) in each tree, and are jittered slightly so they don't overlap. Data from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j.

```
# new data frame at 600 cm for all three species
new_data <- data.frame(
  height_cm = 600,
  ant_species = factor(c("AB", "RRB", "BBR"),
    levels = c("AB", "RRB", "BBR"))
)

# get predicted probabilities
new_data$predicted_prob <- predict(model2, newdata = new_data, type = "response")

new_data
```

	height_cm	ant_species	predicted_prob
1	600	AB	0.2048017
2	600	RRB	0.7407857
3	600	BBR	1.0000000

k.

At the tallest tree height examined (600 cm), the predicted probability of nest occurrence was highest in trees occupied by *C. nigriceps* (0.93) and lowest in trees occupied by *C. sjostedti* (0.47), with *C. mimosae* in between (0.83). As shown in Figure 1, the probability of nest occurrence increased with tree height for all three ant species, but ant species also clearly mattered — birds strongly preferred trees with more aggressive defenders. This likely reflects the fact that *C. nigriceps* and *C. mimosae* protect nests from predators and *C. nigriceps* additionally creates denser canopies through apical bud pruning, providing extra concealment for nesting birds.